

Platform to Identify AI-Generated/AI-Altered Videos & Images and Trace Their Origin Across Social Media

Problem Statement: Development of an AI-Powered Digital Forensic Platform to Detect AI-Generated or Manipulated Images and Videos, Verify Their Authenticity, and Trace Their Origin and Dissemination Across Social Media

Explanation: Generative AI tools can now produce highly realistic synthetic images, videos, and voice clones (deepfakes) that are increasingly difficult to distinguish from authentic media using the naked eye. Bad actors exploit this capability for misinformation campaigns, political manipulation, non-consensual impersonation, financial fraud (including voice-cloned executive scams), and tampering with digital evidence used in investigations. Once manipulated content is uploaded, it spreads rapidly across multiple social media platforms through re-posts, re-uploads, and cross-platform sharing, making it difficult for investigators to identify the original source, the point of first appearance, and the accounts responsible for its amplification. Existing manual verification methods are slow, inconsistent, and cannot keep pace with the volume and sophistication of synthetic content in circulation. There is a pressing need for a unified platform that combines AI-based media forensics with content-provenance verification and cross-platform origin tracing to support law enforcement, journalists, and platform trust-and-safety teams in combating digital misinformation and cyber fraud.

Note: This problem statement concerns building lawful detection and verification tooling. It does not involve creating or improving deepfake-generation techniques; participants should treat any generative components (e.g., for producing labelled training/test samples) as strictly limited to model evaluation, using publicly available research datasets.

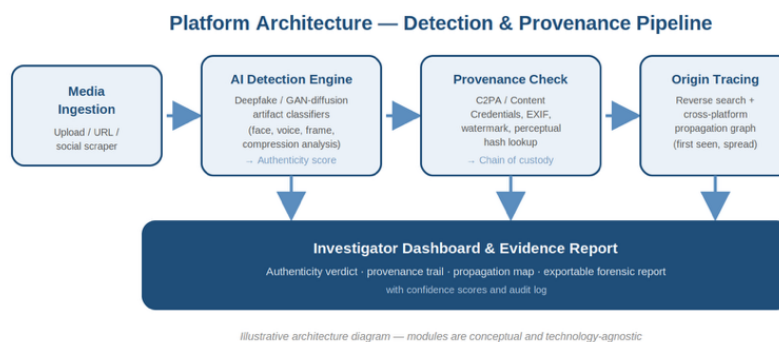


Figure 1: Conceptual pipeline — media ingestion, AI-based detection, provenance/content-credential verification, and cross-platform origin tracing feeding a unified investigator dashboard. (Illustrative architecture diagram.)

Objectives

- To enable rapid, automated detection of AI-generated or manipulated images and videos with an interpretable confidence/authenticity score.
- To verify content authenticity by checking cryptographic provenance signals (e.g., C2PA Content Credentials) and metadata consistency alongside AI-based artifact detection.
- To trace the origin of suspect media — identifying the earliest known appearance, source account, and platform — using reverse search and perceptual-hash matching.
- To visualize the cross-platform propagation of manipulated content, helping investigators understand how and where it spread.
- To support law enforcement, journalists, and platform trust-and-safety teams with exportable, evidence-grade forensic reports.
- To design a scalable system capable of processing high volumes of user-submitted or platform-monitored media with reasonable turnaround time.

Expected Features

The proposed platform may include the following capabilities:

1. Multi-Modal AI Detection Engine

Analyze images, videos, and audio using deep-learning classifiers trained to detect deepfake and GAN/diffusion-generation artifacts, including facial inconsistencies, unnatural blinking, lighting/texture anomalies, and audio-visual lip-sync mismatches.

2. Content Provenance & Metadata Verification

Check for and validate cryptographic content-provenance signals such as C2PA Content Credentials, along with EXIF/metadata consistency, to corroborate or contradict the AI detection result.

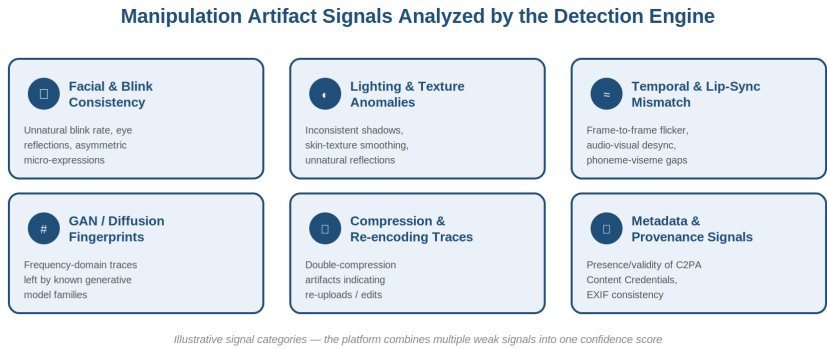


Figure 2: Categories of manipulation-artifact signals the detection engine combines into a single authenticity score. (Illustrative signal taxonomy.)

3. Explainable Authenticity Scoring

Present a clear, interpretable confidence score alongside a visual explanation (e.g., heat-map highlighting suspicious regions/frames) so investigators can understand why content was flagged.

4. Reverse Search & Origin Identification

Use perceptual hashing and reverse image/video search across indexed sources to identify the earliest known appearance of the media and its likely original source.

5. Cross-Platform Propagation Tracing

Track re-uploads, re-posts, and near-duplicate variants of flagged media across multiple social media platforms, mapping the spread over time and by account.

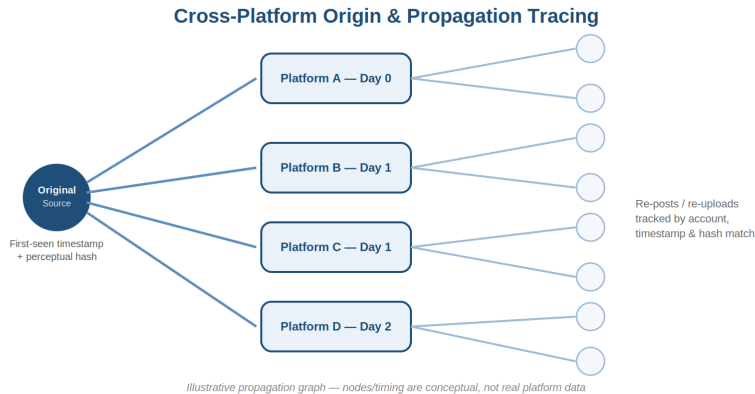


Figure 3: Example propagation graph tracing a manipulated media item from its original source across multiple platforms and accounts. (Illustrative graph — not real platform data.)

6. Investigator Dashboard

Provide a centralized dashboard to submit media for analysis, review authenticity verdicts, explore the origin/propagation graph, and manage ongoing cases.

7. Automated Alerting

Notify analysts when high-confidence manipulated content is detected involving monitored keywords, accounts, or public figures, enabling faster response to emerging misinformation or impersonation campaigns.

8. Evidence-Grade Reporting

Generate structured, exportable forensic reports (detection results, provenance findings, propagation timeline, methodology) suitable to support investigations and legal proceedings, with an audit trail of all analysis steps.

9. Security & Access Control

Ensure secure authentication, role-based access control, and audit logging to protect sensitive case data and maintain chain-of-custody integrity for submitted evidence.

10. Scalability & Model Updatability

Architect the system to process large volumes of media efficiently and to be periodically retrained/updated as new generative-AI techniques emerge, maintaining detection accuracy over time.

Tech Stack:

Participants are encouraged to justify their technology choices based on scalability, security, maintainability, performance, and suitability for the proposed solution rather than adopting any predefined technology stack.

References:

- 1) Tiwari, S. & Kumari, B. (2026). "Development of Digital Forensics Technique Based in AI to Identify Deepfake Video and Image." International Journal of Scientific Research in Science and Technology, 13(3), 01–05. [Link](#)
- 2) Coalition for Content Provenance and Authenticity (C2PA) — Technical Specification for content provenance and Content Credentials. [Link](#)
- 3) C2PA Technical Specification — detailed specification document. [Link](#)